

Mathematical Language

Notation and Conventions

Jungwoo Kim

Department of Mathematics

Pohang University of Science and Technology

Contents

Part I. General Mathematical Language	1
1 General Conventions	1
2 Mathematical Writing	2
Part II. Notation by Mathematical Area	5
3 Set Theory and Logic	5
4 Calculus and Analysis	7
5 Linear Algebra	11
6 Differential Equations	13
7 Combinatorics and Discrete Mathematics	15
8 Probability and Statistics	16
9 Geometry	17
10 Number Theory	19
11 Algebra	20
12 Topology	22
13 Miscellaneous Notation	23

Part I. General Mathematical Language

1 General Conventions

These conventions apply throughout. Widely used standard alternatives may be listed for the same concept in preferred order; the first is used throughout.

Item	Convention
Scalars	Ordinary mathematical italic letters, for example a, b, c, λ, t .
Vectors	Bold lowercase letters, for example $\mathbf{u}, \mathbf{v}, \mathbf{x}$. Unless otherwise stated, vectors are column vectors.
Matrices	Bold uppercase letters, for example $\mathbf{A}, \mathbf{B}, \mathbf{M}$. Parentheses are the preferred matrix delimiters; square brackets are a standard alternative.
Greek vectors and matrices	Bold Greek letters when a Greek symbol denotes a vector or matrix, for example $\boldsymbol{\xi}$ and $\boldsymbol{\Phi}$.
Dimension subscripts	Prefer dimension subscripts when the dimension is readily available and useful; omit them when it is clear, immaterial, or not readily available rather than inferring a dimension solely to supply a subscript.
Number systems	Blackboard bold, for example $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.
Fields	Arbitrary or named fields are denoted by ordinary italic uppercase letters such as F, K, L . Blackboard bold is reserved for established number systems and standard finite-field notation such as $\mathbb{F}_q$.
Sets	Ordinary uppercase letters such as A, B, X, Y , unless a field-specific convention requires otherwise.
Families and collections	Calligraphic letters when useful, for example $\mathcal{F}, \mathcal{T}, \mathcal{B}$.
Named operators	Upright Roman type, for example $\det, \ker, \operatorname{im}, \operatorname{rk}$, and tr .
Large operators	Display size in displayed mathematics and notation tables; sums and products place indices above and below, limits place subscripts below, and integral bounds retain standard right-side placement.
Greek variants	Distinct typographic variants of Greek letters are treated as distinct symbols, not as interchangeable forms of the same symbol. For example, ε and ϵ , and φ and ϕ , are distinct unless explicitly identified otherwise.

Notation	Meaning
$(a_1, \dots, a_n)$	Ordered tuple.
$(a_n), (a_n)_{n \geq 1}$	Sequence; an appropriate natural-number index set may be displayed when useful.
$(a_i)_{i \in I}$	Indexed family over an arbitrary index set I .
(a, b)	Open interval.
$[a, b]$	Closed interval.
$[a, b)$	Left-closed, right-open interval.
$(a, b]$	Left-open, right-closed interval.
$\{a_1, \dots, a_n\}$	Set delimiters.
$\{x \in A \mid P(x)\}$	Set-builder notation.
$\langle u, v \rangle$	Pairing or inner product, according to context.
$ x $	Absolute value, modulus, cardinality, or order, according to context.
$\ x\ $	Norm.
s.t.	Abbreviation for “such that” when naturally required.
$a = b$	Equality.
$a := b$	a is defined to be b .

Notation	Meaning
$a \equiv b$	Identity, equivalence, or congruence, according to context.
$a \cong b$	Isomorphism or congruence, according to context.
$a \approx b$	Approximation.
$f: X \rightarrow Y$	Function or map from X to Y .
$x \mapsto f(x)$	Elementwise rule of a map.
$g \circ f$	Composition: first f , then g .
$f _A, f \upharpoonright_A$	Restriction of f to A .
id_X	Identity map on X .

2 Mathematical Writing

Theorem-like Environments.

Item	Convention
Definition	Precise meaning of a term, object, property, or construction.
Notation	Introduction or declaration of notation.
Theorem	Principal mathematical result.
Proposition	Useful result of more limited scope.
Lemma	Supporting result, usually for a later proof.
Corollary	Result following readily from a preceding result.
Conjecture	Statement proposed to be true but not proved in full generality.
Hypothesis	Named assumption or proposed statement.
Formula	Distinguished computational identity or closed form.
Algorithm	Explicit step-by-step procedure.
Fact	Verified statement not requiring theorem status.
Example	Illustration of a definition, construction, phenomenon, or result.
Counterexample	Example disproving a general assertion or showing a hypothesis is necessary.
Remark	Supplementary context, warning, or qualification.
Problem	Mathematical task intended to be solved.
Open Problem	Problem not known to be solved in the stated form.
Idea	Decisive conceptual insight.
Strategy	Broad problem-solving plan.
Tactic	Local step within a larger strategy.

Proof labels are written as *Proof* and solution labels as *Solution*, each followed by a period.

Notation	Meaning
$\square$	Ending for ordinary proofs and solutions.
$\blacksquare$	Ending for substantial, difficult, or particularly important proofs.
Q.E.D.	Ending for exceptionally difficult, significant, or paper-level proofs.

Enumeration and Subdivision. Enumeration labels are set in upright Roman type. Their form is determined by structural role rather than nesting depth.

Item	Convention
1.1, 1.2, 1.3, ...	Independent child problems or standalone units under a common parent; each is a distinct subproblem that may be stated, solved, and referenced separately.
(1), (2), (3), ...	Ordinary parallel enumeration, such as choices, cases, possibilities, alternatives, examples, or procedural steps; not a subdivision of a single problem or a clause of a formal mathematical statement.
(a), (b), (c), ...	Connected subparts of a single problem or task that share a common statement, setup, or context.
(i), (ii), (iii), ...	Formal mathematical clauses within definitions, theorems, propositions, lemmas, and related statements, such as hypotheses, conditions, properties, conclusions, or equivalent statements.

The distinction between 1.1 and (a) is independence: 1.1 denotes a distinct child problem under a common parent, whereas (a) denotes a connected part of one problem sharing its setup. The distinction between (1) and (i) is function: (1) is used for ordinary parallel enumeration, whereas (i) is reserved for clauses within formal mathematical statements. Nested enumeration preserves these structural roles rather than following a fixed hierarchy of label styles.

Mathematical Shorthand. Typeset abbreviations may be used in typeset mathematical writing as well as handwritten work. Handwritten shorthand is reserved for notes, scratch work, and blackboard writing.

Typeset Abbreviations.

Notation	Meaning
i.e.	“that is.”
i.o.w.	“in other words.”
s.t.	“such that”; preferred when an abbreviation is natural in typeset mathematical writing.
iff	“if and only if.”
w.r.t.	“with respect to.”
const	“constant.”
N.t.	“note that.”
w.l.o.g.	“without loss of generality.”
w.t.s.	“want to show (that).”
e.t.s.	“enough to show (that).”
t.f.a.e.	“the following are equivalent.”
i.t.s.w.	“in the same way.”
o.t.o.h.	“on the other hand.”
IVT	Intermediate Value Theorem.
EVT	Extreme Value Theorem.
MVT	Mean Value Theorem.
FTC	Fundamental Theorem of Calculus.
ODE	Ordinary Differential Equation.
PDE	Partial Differential Equation.
IVP	Initial Value Problem.
BVP	Boundary Value Problem.

Handwritten Shorthand.

Notation	Meaning
Def	“definition.”
Thm	“theorem.”
Prop	“proposition.”
Lem	“lemma.”
Cor	“corollary.”
Pf	“proof.”
Sol	“solution.”
Ex	“example.”
Exer	“exercise.”
Rmk	“remark.”
Sps	“suppose (that).”
cont	“continuous.”
cont.ly	“continuously.”
difb	“differentiable.”
difb.ly	“differentiably.”
difl	“differential.”
bdd	“bounded.”
unif	“uniform.”
unif.ly	“uniformly.”
lin	“linear.”
lin.ly	“linearly.”
dep	“dependent.”
indep	“independent.”
approx	“approximate.”
triv	“trivial.”
fnc	“function.”
seq	“sequence.”
eqn	“equation.”
eqlt	“equality.”
ineqlt	“inequality.”
sys	“system.”
comb	“combination.”
vec	“vector.”
mat	“matrix.”
col	“column.”
pt	“point.”
bdy	“boundary.”
nbhd	“neighborhood.”
itvl	“interval.”
op	“operation.”
num	“number.”
val	“value.”
elmt	“element.”

Notation	Meaning
cw	“clockwise.”
ccw	“counterclockwise.”
w/	“with.”
w/o	“without.”

Mathematical Phrases.

Phrase	Use
We find by inspection that ...	Used when a conclusion follows directly from inspection.

Part II. Notation by Mathematical Area**3 Set Theory and Logic**

Uppercase letters usually denote sets; lowercase letters usually denote their elements.

Notation	Meaning
$\mathbb{N} = \{1, 2, 3, \dots\}$	Positive integers.
$\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$	Nonnegative integers.
$\mathbb{Z}$	Integers.
$\mathbb{Q}$	Rational numbers.
$\mathbb{R}$	Real numbers.
$\mathbb{C}$	Complex numbers.
$\emptyset, \varnothing$	Empty set.
$x \in A$	x is an element of A .
$x \notin A$	x is not an element of A .
$A \subseteq B$	A is a subset of B ; equality is allowed.
$A \subsetneq B$	A is a proper subset of B .
$A \supseteq B$	A is a superset of B ; equality is allowed.
$A \supsetneq B$	A is a proper superset of B .
$A \cup B$	Union.
$A \cap B$	Intersection.
$(A_i)_{i \in I}$	Indexed family of sets.
$\bigcup_{i \in I} A_i$	Union of an indexed family of sets.
$\bigcap_{i \in I} A_i$	Intersection of an indexed family of sets.
$A \setminus B, A - B$	Set difference.
$A \triangle B, A \Delta B$	Symmetric difference; $A \triangle B$ is preferred.
A^c	Complement of A in the ambient set.
$\mathcal{P}(A), 2^A$	Power set.
$A \times B$	Cartesian product.
$A_1 \times \dots \times A_n, A^n$	Finite Cartesian product; A^n is the n -fold Cartesian power of A .
$\prod_{i \in I} A_i$	Cartesian product of an indexed family of sets.
$A \sqcup B, A \dot{\cup} B$	Disjoint union.
$\bigsqcup_{i \in I} A_i$	Disjoint union of an indexed family of sets.

Notation	Meaning
$A \cap B = \emptyset$	A and B are disjoint.
$\{x \in A \mid P(x)\}$	Elements $x \in A$ satisfying $P(x)$.
$R \subseteq X \times Y$	Binary relation between X and Y .
xRy	x is related to y by the relation R .
R^{-1}	Inverse relation of R .
$[x]_R, [x], R[x]$	Equivalence class of x under R ; abbreviated forms are used when the relation is clear.
$x \sim y$	Equivalent elements under an understood equivalence relation.
$A/\sim$	Set of equivalence classes.
$\text{dom } f$	Domain of f .
$\text{codom } f$	Codomain of f .
$f(A)$	Image of A under f .
$f^{-1}(B)$	Preimage of B under f ; this notation does not require f to be invertible.
$\text{im } f$	Image of the map f .
$f^{-1}: Y \rightarrow X$	Inverse function of a bijection $f: X \rightarrow Y$.
$f: X \hookrightarrow Y$	Injective map.
$f: X \twoheadrightarrow Y$	Surjective map.
$f: X \xrightarrow{\sim} Y$	Bijjective map.
$f: X \xrightarrow{\sim} Y$	Isomorphism; for sets, equivalently a bijection.
$\mathbb{1}_A(x)$	Indicator function of A .
$\forall x \in A$	For every $x \in A$.
$\exists x \in A$	There exists $x \in A$.
$\exists! x \in A$	There exists a unique $x \in A$.
$\nexists x \in A$	There does not exist $x \in A$.
$P \Rightarrow Q$	P implies Q in ordinary mathematical writing.
$P \Leftarrow Q$	Q implies P in ordinary mathematical writing.
$P \nRightarrow Q$	P does not imply Q .
$P \Leftrightarrow Q$	P if and only if Q in ordinary mathematical writing.
$P \wedge Q$	Logical conjunction.
$P \vee Q$	Logical disjunction.
$\neg P$	Logical negation.
$\top, \perp$	Truth and falsehood constants, respectively, in formal logic.
$\varphi \rightarrow \psi, \varphi \leftrightarrow \psi$	Conditional and biconditional connectives in formal logic.
$\mathcal{M} \models \varphi, \mathcal{M} \not\models \varphi$	$\mathcal{M}$ satisfies, or does not satisfy, the formula φ .
$\Gamma \models \varphi, \Gamma \not\models \varphi$	φ is, or is not, a semantic consequence of Γ .
$\Gamma \vdash \varphi, \Gamma \not\vdash \varphi$	φ is, or is not, syntactically derivable from Γ .
$\bot$	Contradiction reached in a proof or argument; distinct from the formal falsehood constant $\perp$.
$\therefore$	Therefore; indicates a conclusion from preceding statements.
$\because$	Because; indicates a reason or justification.
$ A , \#A$	Cardinality; $ A $ is preferred, while $\#A$ is a common alternative, especially when emphasizing the number of elements.
$\aleph_0$	Cardinality of a countably infinite set.
$\mathfrak{c}$	Cardinality of the continuum.

4 Calculus and Analysis

The differential d is upright; ε is the default small positive quantity in analysis. Function spaces take their domains in parentheses, for example $C([a, b])$, $\mathcal{R}([a, b])$, and $L^p(\Omega)$.

Notation	Meaning
$\lim_{x \rightarrow a} f(x)$	Limit as x approaches a .
$\lim_{x \rightarrow a^-} f(x)$	Left-hand limit as x approaches a .
$\lim_{x \rightarrow a^+} f(x)$	Right-hand limit as x approaches a .
$\lim_{x \rightarrow \infty} f(x), \lim_{x \rightarrow -\infty} f(x)$	Limits as x tends to positive or negative infinity.
$\lim_{x \rightarrow a} f(x) = \infty,$ $\lim_{x \rightarrow a} f(x) = -\infty$	Infinite limits.
$\min A$	Minimum of A .
$\max A$	Maximum of A .
$\inf A$	Infimum of A .
$\sup A$	Supremum of A .
$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$	Extended real line.
$f'(a)$	First derivative of f at a .
$f^{(n)}(a)$	n th derivative of f at a .
$\frac{dy}{dx}$	Derivative of y with respect to x .
$\frac{d^n y}{dx^n}$	n th derivative of y with respect to x in Leibniz notation.
$\frac{\partial f}{\partial x_i}, \partial_i f, f_{x_i}$	Partial derivative with respect to x_i .
$\frac{\partial^2 f}{\partial x_i \partial x_j}, \partial_{ij} f, f_{x_i x_j}$	Second or mixed partial derivative.
$D_{\mathbf{v}} f(\mathbf{a}), \partial_{\mathbf{v}} f(\mathbf{a})$	Directional derivative along the vector $\mathbf{v}$, which need not be a unit vector: $D_{\mathbf{v}} f(\mathbf{a}) = \left. \frac{d}{dt} f(\mathbf{a} + t\mathbf{v}) \right _{t=0} = Df(\mathbf{a})[\mathbf{v}] = \nabla f(\mathbf{a}) \cdot \mathbf{v}$ when f is differentiable. For the unit direction determined by $\mathbf{v} \neq \mathbf{0}$, use $\mathbf{u} = \mathbf{v}/\ \mathbf{v}\ $.
$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$	Total differential of a scalar function of n variables.
$\int f(x) dx$	Indefinite integral; a family of antiderivatives, with an additive constant understood.
$\int_a^b f(x) dx$	Definite integral over $[a, b]$.
$\int_a^\infty f(x) dx,$ $\int_{-\infty}^b f(x) dx, \int_{-\infty}^\infty f(x) dx$	Improper integrals over unbounded intervals.
$\int_a^b f d\alpha$	Riemann–Stieltjes integral of f with respect to α .
$P = \{x_0, \dots, x_n\},$ $a = x_0 < \dots < x_n = b$	Partition of the interval $[a, b]$.
$\sum_{i=1}^n f(\xi_i) \Delta x_i,$ $\Delta x_i = x_i - x_{i-1}$	Tagged Riemann sum, with $\xi_i \in [x_{i-1}, x_i]$.
$\iint_D f dA$	Double integral over D .

Notation	Meaning
$\iiint_{\Omega} f \, dV$	Triple integral over Ω .
$\int_C f(z) \, dz$	Contour integral along an oriented contour C .
$\oint_C f(z) \, dz$	Contour integral over a closed contour C .
$-C$	The contour C with the opposite orientation.
γ^{-1}	Reversed parametrized path γ , when unambiguous; not a functional inverse.
PV, P.V.	Principal value; in particular, the Cauchy principal value of a singular or improper integral.
$\sum_{k=1}^n a_k$	Finite sum.
$\sum_{k=1}^{\infty} a_k$	Infinite series.
$\prod_{k=1}^n a_k$	Finite product.
$\sum_{n=0}^{\infty} a_n (x-a)^n$	Power series centered at a .
$\sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$	n th Taylor polynomial of f about a ; no single letter for the polynomial is imposed globally.
$e^x, \exp x$	Exponential function.
$\ln x, \log_e x$	Logarithm to base e .
$\lg x, \log_{10} x$	Logarithm to base 10.
$\text{lb } x, \log_2 x$	Logarithm to base 2.
$\log_a x$	Logarithm to base a .
$\sin x$	Sine.
$\cos x$	Cosine.
$\tan x$	Tangent.
$\csc x$	Cosecant.
$\sec x$	Secant.
$\cot x$	Cotangent.
$\arcsin x, \sin^{-1} x$	Principal inverse sine, $[-1, 1] \rightarrow [-\pi/2, \pi/2]$.
$\arccos x, \cos^{-1} x$	Principal inverse cosine, $[-1, 1] \rightarrow [0, \pi]$.
$\arctan x, \tan^{-1} x$	Principal inverse tangent, $\mathbb{R} \rightarrow (-\pi/2, \pi/2)$.
$\text{arccsc } x, \csc^{-1} x$	Principal inverse cosecant, defined by $\text{arccsc } x = \arcsin(1/x)$ for $ x \geq 1$; its range is $[-\pi/2, 0) \cup (0, \pi/2]$.
$\text{arcsec } x, \sec^{-1} x$	Principal inverse secant, defined by $\text{arcsec } x = \arccos(1/x)$ for $ x \geq 1$; its range is $[0, \pi/2) \cup (\pi/2, \pi]$.
$\text{arccot } x, \cot^{-1} x$	Principal real inverse cotangent, $\mathbb{R} \rightarrow (0, \pi)$.
$\sinh x$	Hyperbolic sine.
$\cosh x$	Hyperbolic cosine.
$\tanh x$	Hyperbolic tangent.
$\text{csch } x$	Hyperbolic cosecant.
$\text{sech } x$	Hyperbolic secant.
$\coth x$	Hyperbolic cotangent.

Notation	Meaning
$\operatorname{arcsinh} x, \sinh^{-1} x$	Principal real inverse hyperbolic sine, $\mathbb{R} \rightarrow \mathbb{R}$.
$\operatorname{arccosh} x, \cosh^{-1} x$	Principal real inverse hyperbolic cosine, $[1, \infty) \rightarrow [0, \infty)$.
$\operatorname{arctanh} x, \tanh^{-1} x$	Principal real inverse hyperbolic tangent, $(-1, 1) \rightarrow \mathbb{R}$.
$\operatorname{arccsch} x, \operatorname{csch}^{-1} x$	Principal real inverse hyperbolic cosecant, defined by $\operatorname{arccsch} x = \operatorname{arcsinh}(1/x)$ on $\mathbb{R} \setminus \{0\}$; its range is $\mathbb{R} \setminus \{0\}$.
$\operatorname{arcsech} x, \operatorname{sech}^{-1} x$	Principal real inverse hyperbolic secant, defined by $\operatorname{arcsech} x = \operatorname{arccosh}(1/x)$ on $(0, 1]$; its range is $[0, \infty)$.
$\operatorname{arcoth} x, \operatorname{coth}^{-1} x$	Principal real inverse hyperbolic cotangent, defined by $\operatorname{arcoth} x = \operatorname{arctanh}(1/x)$ for $ x > 1$; its range is $\mathbb{R} \setminus \{0\}$.
$\Gamma(z)$	Gamma function.
$B(x, y)$	Beta function.
$\zeta(s)$	Riemann zeta function.
$\mathbf{J}_{\mathbf{F}}(\mathbf{a}), D\mathbf{F}(\mathbf{a})$	Jacobian matrix of $\mathbf{F} = (F_1, \dots, F_m): \mathbb{R}^n \rightarrow \mathbb{R}^m$ at $\mathbf{a}$, with $\mathbf{J}_{\mathbf{F}}(\mathbf{a}) = \left(\frac{\partial F_i}{\partial x_j}(\mathbf{a}) \right)_{i,j}$. The notation $D\mathbf{F}(\mathbf{a})$ denotes the total derivative, identified with this matrix in standard coordinates.
$\det \mathbf{J}_{\mathbf{F}}(\mathbf{a}), \det D\mathbf{F}(\mathbf{a}), \frac{\partial(F_1, \dots, F_n)}{\partial(x_1, \dots, x_n)} \Big _{\mathbf{a}}$	Jacobian determinant of $\mathbf{F}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ at $\mathbf{a}$, in preferred order.
$\operatorname{Hess} f$	Hessian matrix of f .
$\nabla f, \operatorname{grad} f$	Gradient of f .
$\nabla \cdot \mathbf{F}, \operatorname{div} \mathbf{F}$	Divergence of $\mathbf{F}$.
$\nabla \times \mathbf{F}, \operatorname{curl} \mathbf{F}$	Curl of $\mathbf{F}$.
Δf	Laplacian of f .
$\mathbf{r}(t)$	Parametrized curve.
ds	Element of arc length.
$\int_C f \, ds$	Scalar line integral along C with respect to arc length.
$\int_C \mathbf{F} \cdot d\mathbf{r}$	Line integral of the vector field $\mathbf{F}$ along C .
$\iint_S f \, dS$	Scalar surface integral over S .
$\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$	Flux integral through S .
$x_n \rightarrow x, \lim_{n \rightarrow \infty} x_n = x$	Convergence of a sequence.
(x_{n_k})	Subsequence of the sequence (x_n) .
$a_n \nearrow L$	Monotone increasing convergence to L .
$a_n \searrow L$	Monotone decreasing convergence to L .
$\limsup a_n$	Limit superior.
$\liminf a_n$	Limit inferior.
$f(x) \sim g(x)$	Asymptotic equivalence in the stated limiting regime, typically meaning $f(x)/g(x) \rightarrow 1$.
$f_n \rightarrow f, f_n \xrightarrow{\text{ptw}} f$	Pointwise convergence of functions on a common domain by default when no ambient topology, metric, or norm is specified. If such a structure is specified, the ordinary arrow $\rightarrow$ refers to convergence in that structure.
$f_n \rightrightarrows f, f_n \xrightarrow{\text{unif}} f$	Uniform convergence.
$f_n \xrightarrow{\text{a.e.}} f$	Convergence almost everywhere.
$\mu(E)$	Measure of a measurable set E .

Notation	Meaning
$\int_E f \, d\mu$	Integral of f over E with respect to the measure μ .
$f = g$ a.e., $f = g$ μ -a.e.	Equality almost everywhere, with the measure omitted when understood.
$\operatorname{ess\,sup}_{x \in E} f(x)$	Essential supremum of f on E .
$\operatorname{ess\,inf}_{x \in E} f(x)$	Essential infimum of f on E .
$C(\Omega)$, $C^0(\Omega)$	Continuous functions on Ω .
$C^k(\Omega)$	k times continuously differentiable functions on Ω .
$C^\infty(\Omega)$	Smooth functions on Ω .
$\operatorname{supp} f$	Support of f , the closure of the set on which f is nonzero.
$C_c(\Omega)$, $C_c^\infty(\Omega)$	Continuous and smooth functions on Ω with compact support, respectively.
$\mathcal{R}([a, b])$	Riemann-integrable functions on $[a, b]$.
$L^p(\Omega)$, $L^p(X, \mu)$	Lebesgue L^p spaces; the measure is displayed when it is not understood.
$L^\infty(\Omega)$, $L^\infty(X, \mu)$	Lebesgue spaces of essentially bounded functions.
ℓ^p	Space of scalar sequences (x_n) with $\sum_{n=1}^\infty x_n ^p < \infty$, for $1 \leq p < \infty$.
ℓ^∞	Space of bounded scalar sequences.
$\ f\ _\infty$	Supremum norm $\sup_{x \in \Omega} f(x) $ for an actual bounded function, especially in bounded- or continuous-function spaces.
$\ f\ _{L^p(\Omega)}$, $\ f\ _p$	L^p -norm $\left(\int_\Omega f(x) ^p \, dx \right)^{1/p}$ for $1 \leq p < \infty$.
$\ f\ _{L^\infty(\Omega)}$	L^∞ -norm $\operatorname{ess\,sup}_{x \in \Omega} f(x) $; this notation is preferred when distinguishing the essential-supremum norm from an ordinary supremum norm.
$\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$	Multi-index; α is not treated as a vector for typographic purposes.
$ \alpha = \alpha_1 + \dots + \alpha_n$	Order and factorial of a multi-index.
$\alpha! = \alpha_1! \dots \alpha_n!$	Order and factorial of a multi-index.
$\mathbf{x}^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$	Multi-index monomial for $\mathbf{x} = (x_1, \dots, x_n)$.
$D^\alpha f$, $\partial^\alpha f$	Mixed partial derivative specified by the multi-index α .
$B_r(x)$	Open ball of radius r centered at x .
$\overline{B}_r(x)$	Closed ball of radius r centered at x .
$\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$	Extended complex plane, equivalently the Riemann sphere.
$z = x + iy$	Complex number with real part x and imaginary part y .
$\operatorname{Re} z$, $\Re z$	Real part of z .
$\operatorname{Im} z$, $\Im z$	Imaginary part of z ; distinct from lowercase im for image.
$\bar{z}$	Complex conjugate.
$ z $	Complex modulus.
$\arg z$	Argument of a nonzero complex number z ; its values differ by integer multiples of 2π .
$\operatorname{Arg} z$	Principal argument of $z \neq 0$, with $-\pi < \operatorname{Arg} z \leq \pi$.
$\operatorname{Ln} z$	General multivalued complex logarithm for $z \neq 0$: $\operatorname{Ln} z = \{\ln z + i(\operatorname{Arg} z + 2\pi k) \mid k \in \mathbb{Z}\}$.
$\ln z = \ln z + i \operatorname{Arg} z$	Principal value of the complex logarithm for $z \neq 0$. On $\mathbb{C} \setminus (-\infty, 0]$, it is the analytic principal branch, with $-\pi < \operatorname{Arg} z < \pi$.
$\operatorname{Res}(f; a)$, $\operatorname{Res}_{z=a} f(z)$	Residue of f at a finite point a .

Notation	Meaning
$\text{Res}(f; \infty)$, $\text{Res}_{z=\infty} f(z)$	Residue of f at infinity.

5 Linear Algebra

Notation	Meaning
$M_{m \times n}(F)$, $F^{m \times n}$	Space of $m \times n$ matrices over the field F ; $M_{m \times n}(F)$ is preferred.
$M_n(F)$	Space (and matrix algebra) of $n \times n$ matrices over F .
$\mathbf{A} = (a_{ij})$, $\mathbf{A} = [a_{ij}]$	Matrix with (i, j) -entry a_{ij} ; parentheses are preferred.
$\mathbf{A}^\top$	Transpose of $\mathbf{A}$.
$\mathbf{A}^H$	Conjugate transpose of $\mathbf{A}$.
$\mathbf{A}^{-1}$	Inverse of $\mathbf{A}$.
$\mathbf{I}_n$, $\mathbf{I}$	Identity matrix; the dimension subscript is preferred when readily available and useful.
$\mathbf{0}$	Zero vector or zero matrix, with dimension understood.
$\mathbf{e}_i$	i th standard basis vector.
$\mathbf{1}$	All-ones vector.
$\mathbf{D}$	Diagonal matrix when a local name is required.
$\text{diag}(d_1, \dots, d_n)$	Diagonal matrix with entries $d_1, \dots, d_n$.
$\mathbf{P}$	Projection matrix when a local name is required.
$\det(\mathbf{A})$, $ \mathbf{A} $	Determinant; $\det(\mathbf{A})$ is preferred.
$\text{rk}(\mathbf{A})$, $\text{rank}(\mathbf{A})$	Rank; $\text{rk}(\mathbf{A})$ is preferred.
$\text{tr}(\mathbf{A})$	Trace.
$\text{adj}(\mathbf{A})$	Adjugate matrix.
M_{ij}	Minor of the (i, j) -entry, when the matrix is understood.
$C_{ij} = (-1)^{i+j} M_{ij}$	Cofactor of the (i, j) -entry.
$\text{Col}(\mathbf{A})$	Column space.
$\text{Row}(\mathbf{A})$	Row space.
$\ker \mathbf{A}$, $\text{Null}(\mathbf{A})$	Null space of $\mathbf{A}$; $\ker \mathbf{A}$ is preferred, while $\text{Null}(\mathbf{A})$ is a common matrix-oriented alternative.
V, W	Vector spaces, usually over an understood field F .
$\dim V$	Dimension of V .
$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$	Span of the listed vectors.
$U \oplus V$	Direct sum.
$\ker T$	Kernel of a linear map T .
$\text{im } T$	Image of a linear map T .
$\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$	Ordered basis.
$[\mathbf{v}]_{\mathcal{B}}$	Coordinate vector of $\mathbf{v}$ with respect to the ordered basis $\mathcal{B}$.
$[T]_{\mathcal{C} \leftarrow \mathcal{B}}$	Matrix of $T: V \rightarrow W$ with respect to domain basis $\mathcal{B}$ and codomain basis $\mathcal{C}$; thus $[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{v}]_{\mathcal{B}}$.
$[T]_{\mathcal{B}}$	Matrix of an endomorphism $T: V \rightarrow V$ when the same basis $\mathcal{B}$ is used in the domain and codomain.
$\mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}}$	Change-of-coordinate matrix from $\mathcal{B}$ -coordinates to $\mathcal{C}$ -coordinates; $[\mathbf{v}]_{\mathcal{C}} = \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}} [\mathbf{v}]_{\mathcal{B}}$.
V^*	Dual space of V .
$\langle \mathbf{u}, \mathbf{v} \rangle$, $(\mathbf{u}, \mathbf{v})$	Inner product; angle brackets are preferred. Over complex spaces, the inner product is conjugate-linear in the first argument and linear in the second.

Notation	Meaning
$\mathbf{u} \cdot \mathbf{v}, \mathbf{u}^\top \mathbf{v}$	Standard Euclidean inner product on $\mathbb{R}^n$.
$\mathbf{u}^H \mathbf{v}$	Standard Hermitian inner product on $\mathbb{C}^n$.
$\ \mathbf{v}\ $	Norm of $\mathbf{v}$.
$\mathbf{u} \perp \mathbf{v}$	Orthogonality.
$U^\perp$	Orthogonal complement of the subspace U .
$\text{proj}_U \mathbf{x}$	Orthogonal projection of $\mathbf{x}$ onto the subspace U .
$\text{proj}_{\mathbf{v}} \mathbf{x}$	Orthogonal projection of $\mathbf{x}$ onto $\text{span}(\mathbf{v})$.
$\mathbf{u} \times \mathbf{v}$	Cross product in $\mathbb{R}^3$.
$\mathbf{A}^\top = \mathbf{A}$	Real symmetric matrix.
$\mathbf{A}^\top = -\mathbf{A}$	Real skew-symmetric matrix.
$\mathbf{A}^H = \mathbf{A}$	Hermitian matrix.
$\mathbf{A}^H = -\mathbf{A}$	Skew-Hermitian matrix.
$\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$	Real orthogonal matrix $\mathbf{Q}$.
$\mathbf{U}^H \mathbf{U} = \mathbf{I}$	Unitary matrix $\mathbf{U}$.
$\mathbf{A} \mathbf{A}^H = \mathbf{A}^H \mathbf{A}$	Normal matrix.
$\mathbf{P}^2 = \mathbf{P}$	Projection matrix.
$\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^\top$	Orthogonal projection matrix over $\mathbb{R}$.
$\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^H$	Orthogonal projection matrix over $\mathbb{C}$.
$\chi_{\mathbf{A}}(t), p_{\mathbf{A}}(t)$	Characteristic polynomial of $\mathbf{A}$, with $\chi_{\mathbf{A}}(t) = \det(t\mathbf{I} - \mathbf{A})$; $\chi_{\mathbf{A}}$ is preferred.
$m_{\mathbf{A}}(t), \mu_{\mathbf{A}}(t)$	Minimal polynomial of $\mathbf{A}$; $m_{\mathbf{A}}$ is preferred.
λ, λ_i	Eigenvalue of $\mathbf{A}$.
$\mathbf{v}, \mathbf{v}_i$	Eigenvector of $\mathbf{A}$.
$E_\lambda(\mathbf{A}), V_\lambda(\mathbf{A})$	Eigenspace associated with λ ; $E_\lambda(\mathbf{A}) = \ker(\mathbf{A} - \lambda\mathbf{I})$.
$\sigma(\mathbf{A}), \text{spec}(\mathbf{A})$	Spectrum of $\mathbf{A}$; $\sigma(\mathbf{A})$ is preferred.
$\rho(\mathbf{A})$	Spectral radius.
$\mathbf{P}_\lambda$	Spectral projection associated with the eigenvalue λ , when defined.
$\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$	Diagonalization of a diagonalizable matrix.
$\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^\top$	Orthogonal diagonalization of a real symmetric matrix, with $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$.
$\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^H$	Unitary diagonalization of a complex normal matrix, with $\mathbf{U}^H \mathbf{U} = \mathbf{I}$.
$\mathbf{A} = \sum_{\lambda} \lambda \mathbf{P}_\lambda$	Spectral decomposition of a diagonalizable matrix, with the sum over distinct eigenvalues.
$e^{\mathbf{A}}, \exp(\mathbf{A})$	Matrix exponential; $e^{t\mathbf{A}}$ is the standard time-dependent form.
$\mathbf{A} = \mathbf{L} \mathbf{U}$	LU factorization, with $\mathbf{L}$ lower triangular and $\mathbf{U}$ upper triangular.
$\mathbf{A} = \mathbf{Q} \mathbf{R}$	QR factorization, with $\mathbf{Q}$ orthogonal or unitary and $\mathbf{R}$ upper triangular.
$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H$	Singular value decomposition (SVD); over $\mathbb{R}$, $\mathbf{V}^H = \mathbf{V}^\top$.
$\sigma_i(\mathbf{A})$	i th singular value of $\mathbf{A}$.
$\mathbf{A}^\dagger$	Moore–Penrose pseudoinverse of $\mathbf{A}$.
$\mathbf{A} \succ 0$	Positive definite Hermitian matrix (real symmetric in the real case).
$\mathbf{A} \succeq 0$	Positive semidefinite Hermitian matrix.
$\mathbf{A} \prec 0$	Negative definite Hermitian matrix.
$\mathbf{A} \preceq 0$	Negative semidefinite Hermitian matrix.
$\mathbf{A} \succeq \mathbf{B}$	Löwner order: $\mathbf{A} - \mathbf{B} \succeq 0$, for Hermitian matrices of the same size.
$\mathbf{A} = \mathbf{L} \mathbf{L}^H$	Cholesky factorization of a positive definite Hermitian matrix; over $\mathbb{R}$, $\mathbf{L}^H = \mathbf{L}^\top$.

6 Differential Equations

Notation	Meaning
y'	First derivative with respect to the understood independent variable.
y''	Second derivative.
$\dot{x}$	First derivative of x with respect to time.
$\ddot{x}$	Second derivative of x with respect to time.
$F(t, y, y', \dots, y^{(n)}) = 0$	General n th-order ordinary differential equation.
$y^{(n)} = f(t, y, \dots, y^{(n-1)})$	n th-order ordinary differential equation in normal form.
$y(t_0) = y_0$	Initial condition for a scalar differential equation.
$y' + p(t)y = g(t)$	Standard form of a first-order linear differential equation.
$\mu(t)$	Integrating factor; for $y' + p(t)y = g(t)$, one may take $\mu(t) = \exp\left(\int p(t) \, dt\right)$ up to a nonzero constant factor.
$M(x, y) \, dx + N(x, y) \, dy = 0$	Differential form of a first-order differential equation.
M_x, M_y, N_x, N_y	Subscript notation for the corresponding partial derivatives.
$L[y]$	Linear differential operator applied to y ; $\mathcal{L}$ is reserved here for the Laplace transform.
$L[y] = \sum_{k=0}^n a_k(t)y^{(k)}$	General n th-order linear differential operator, with $a_n \neq 0$.
$L[y] = 0, \, L[y] = g$	Homogeneous and nonhomogeneous linear differential equations, respectively.
$y_h, \, y_c$	Homogeneous or complementary part of a solution; y_h is preferred.
y_p	Particular solution.
$c_1, c_2, \dots$	Arbitrary constants in a general solution.
$\sum_{k=0}^n a_k r^k = 0$	Characteristic equation for a constant-coefficient homogeneous linear differential equation.
$r_1, \dots, r_n$	Characteristic roots; multiplicities are counted when relevant.
$W(y_1, \dots, y_n)(t),$ $W[y_1, \dots, y_n](t)$	Wronskian of $y_1, \dots, y_n$; parentheses are preferred.
$y = \sum_{n=0}^{\infty} a_n(x - x_0)^n$	Power-series form about the ordinary point x_0 .
$y = \sum_{n=0}^{\infty} a_n(x - x_0)^{n+r}$	Frobenius-series form about a regular singular point x_0 ; r denotes an indicial exponent.
$\mathbf{x}' = \mathbf{f}(t, \mathbf{x})$	General first-order system.
$\mathbf{x}(t_0) = \mathbf{x}_0$	Initial condition for a first-order system.
$\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t),$ $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$	Linear first-order system in state-vector form.
$\mathbf{x}' = \mathbf{f}(\mathbf{x})$	Autonomous first-order system.
$\mathbf{x}_e$	Equilibrium point; $\mathbf{f}(\mathbf{x}_e) = \mathbf{0}$.
$\Phi(t)$	Fundamental matrix of a linear system.
$\mathcal{L}\{f(t)\}(s),$ $\mathcal{L}[f](s)$	Laplace transform of f .
$\mathcal{L}^{-1}\{F(s)\}(t),$ $\mathcal{L}^{-1}[F](t)$	Inverse Laplace transform of F .
$(f * g)(t)$	One-sided convolution, $(f * g)(t) := \int_0^t f(\tau)g(t - \tau) \, d\tau$.

Notation	Meaning
$H(t-a), u(t-a)$	Heaviside or unit-step function shifted to $t=a$; H is preferred, and its value at the jump is specified when relevant.
$\delta(t), \delta(t-a)$	Dirac delta distribution, or unit impulse, centered at 0 or shifted to $t=a$.
$t_n = t_0 + nh$	Uniform numerical grid with step size $h > 0$.
$y_n \approx y(t_n)$	Numerical approximation to the exact solution at t_n .
$u_x, u_t, u_{xx}, u_{xy}, u_{tt}, \dots$	Subscript notation for partial derivatives of a scalar function u .
$\partial_{\mathbf{n}}u$	Normal derivative, $\partial_{\mathbf{n}}u := \nabla u \cdot \mathbf{n}$, for a unit normal vector $\mathbf{n}$.
$u _{\partial\Omega} = g$	Dirichlet boundary condition on $\partial\Omega$.
$\partial_{\mathbf{n}}u _{\partial\Omega} = g$	Neumann boundary condition on $\partial\Omega$.
$\alpha u + \beta \partial_{\mathbf{n}}u = g$	Robin boundary condition on $\partial\Omega$.
a_n, b_n	Fourier coefficients for the real $2L$ -periodic convention
	$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right).$
a_n	$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \text{ for } n \geq 0.$
b_n	$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \text{ for } n \geq 1.$
$\hat{f}(\boldsymbol{\xi}), \mathcal{F}f(\boldsymbol{\xi})$	Fourier transform under the normalization $\hat{f}(\boldsymbol{\xi}) := \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} d\mathbf{x}; \text{ inversion uses}$ $f(\mathbf{x}) = \int_{\mathbb{R}^n} \hat{f}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} d\boldsymbol{\xi} \text{ whenever applicable.}$
$-(py')' + qy = \lambda wy$	Sturm–Liouville eigenvalue equation, where p, q , and w are functions of x .
λ_n, ϕ_n	Sturm–Liouville eigenvalue and corresponding eigenfunction.
$\langle f, g \rangle_w$	Weighted inner product, $\langle f, g \rangle_w := \int_a^b \overline{f(x)} g(x) w(x) dx$.
$\langle \phi_m, \phi_n \rangle_w = \delta_{mn}$	Weighted orthonormality of eigenfunctions.
$G(x, \xi), G(\mathbf{x}, \boldsymbol{\xi})$	Green's function; arguments are scalar or vector as appropriate.
$\text{Ai}(x), \text{Bi}(x)$	Airy functions.
$J_{\nu}(x)$	Bessel function of the first kind of order ν .
$Y_{\nu}(x), N_{\nu}(x)$	Bessel function of the second kind of order ν ; Y_{ν} is preferred.
$I_{\nu}(x), K_{\nu}(x)$	Modified Bessel functions of the first and second kinds.
$H_{\nu}^{(1)}(x), H_{\nu}^{(2)}(x)$	Hankel functions of the first and second kinds.
$P_n(x)$	Legendre polynomial of degree n .
$P_{\nu}(x), Q_{\nu}(x)$	Legendre functions of the first and second kinds.
$P_{\ell}^m(x)$	Associated Legendre function for integer degree ℓ and order m .
$H_n(x), \text{He}_n(x)$	Physicists' and probabilists' Hermite polynomials, respectively.
$L_n^{(\alpha)}(x), L_n(x)$	Generalized and ordinary Laguerre polynomials; $L_n(x) = L_n^{(0)}(x)$.
$T_n(x), U_n(x)$	Chebyshev polynomials of the first and second kinds.
$P_n^{(\alpha, \beta)}(x)$	Jacobi polynomial.
$C_n^{(\lambda)}(x)$	Gegenbauer (ultraspherical) polynomial.
$\text{erf}(x), \text{erfc}(x)$	Error function and complementary error function.
${}_2F_1(a, b; c; z)$	Gauss hypergeometric function.

7 Combinatorics and Discrete Mathematics

Notation	Meaning
$\binom{n}{r}, {}_nC_r$	Number of r -element subsets of an n -element set.
$\left(\binom{n}{r}\right), {}_nH_r$	r -combinations with repetition from n types; equals $\binom{n+r-1}{r}$.
$n!$	Factorial.
$n^{\underline{k}}$	Falling factorial, $n(n-1)\cdots(n-k+1)$.
$n^{\overline{k}}$	Rising factorial, $n(n+1)\cdots(n+k-1)$.
$(a)_k$	Rising Pochhammer symbol $a(a+1)\cdots(a+k-1)$ in special-functions contexts; it is not used here for the falling factorial.
$O(g(n))$	Big-O upper asymptotic bound.
$o(g(n))$	Little-o strict asymptotic bound.
$\Theta(g(n))$	Two-sided asymptotic order.
$\Omega(g(n))$	Asymptotic lower bound.
$(X, \preceq)$	Partially ordered set (poset).
$\prec, \succeq, \succ$	Strict and reverse order relations associated with $\preceq$.
$\leq_{\text{lex}}$	Lexicographic order.
$x \prec y$	Cover relation: y covers x .
B_n	Boolean lattice (Boolean poset) on an n -element ground set.
$G = (V, E)$	Graph with vertex set V and edge set E .
$V(G)$	Vertex set of G .
$E(G)$	Edge set of G .
$N(v), \Gamma(v)$	Neighborhood of vertex v .
$\deg_G(v), \deg(v)$	Degree of vertex v ; the graph subscript may be omitted when clear.
$\deg^+(v), \deg^-(v)$	Out-degree and in-degree of v in a directed graph.
$d_G(u, v), d(u, v)$	Graph distance between vertices u and v .
K_n	Complete graph on n vertices.
$K_{m,n}$	Complete bipartite graph with parts of sizes m and n .
P_n	Path graph on n vertices.
C_n	Cycle graph on n vertices.
$\mathbf{A}(G), \mathbf{A}_G$	Adjacency matrix of G .
$\mathbf{L}(G), \mathbf{L}_G$	Graph Laplacian, usually $\mathbf{D}(G) - \mathbf{A}(G)$.
$G - e$	Graph obtained by deleting the edge e .
$G/e, G : e$	Graph obtained by contracting the edge e ; G/e is the preferred notation.
$T(G), \tau(G)$	Number of spanning trees of G ; $T(G)$ is preferred here to avoid conflict with the standard number-theoretic $\tau(n)$.
$r(k, \ell), r(k)$	Ramsey numbers; $r(k) = r(k, k)$.
$A(x) = \sum_{n=0}^{\infty} a_n x^n$	Ordinary generating function of the sequence $(a_n)_{n \geq 0}$.

8 Probability and Statistics

Probability and expectation are written in blackboard bold by default. Parameter symbols are local to the relevant context: for gamma distributions, β denotes a rate parameter and θ a scale parameter, so $\theta = 1/\beta$, while in hypothesis testing β denotes a Type II error probability. The geometric distribution counts the number of trials up to and including the first success.

Notation	Meaning
$\mathbb{P}(A)$, $P(A)$	Probability of event A .
$\mathbb{P}(A \mid B)$, $P(A \mid B)$	Conditional probability of A given B .
$X \perp\!\!\!\perp Y$	X and Y are independent.
$X \perp\!\!\!\perp Y \mid Z$	X and Y are conditionally independent given Z .
$p_X(x) = \mathbb{P}(X = x)$	Probability mass function (pmf) of a discrete random variable X .
$f_X(x)$	Probability density function (pdf) of a continuous random variable X .
$F_X(x) = \mathbb{P}(X \leq x)$	Cumulative distribution function (cdf) of X .
$p_{X,Y}(x, y)$, $f_{X,Y}(x, y)$	Joint pmf or joint pdf of (X, Y) , according to context.
$p_{X Y}(x \mid y)$, $f_{X Y}(x \mid y)$	Conditional pmf or conditional pdf of X given $Y = y$, according to context.
$F^{-1}(p)$	Generalized inverse of a cdf F , $F^{-1}(p) = \inf\{x \in \mathbb{R} \mid F(x) \geq p\}$; the p -quantile, for $0 < p < 1$.
$\mathbb{E}[X]$, $E[X]$	Expectation of random variable X .
$\mathbb{E}[X \mid Y]$, $E[X \mid Y]$	Conditional expectation of X given Y .
$\text{Var}(X)$	Variance of X .
$\text{Cov}(X, Y)$	Covariance of X and Y .
$\rho_{X,Y}$	Population correlation coefficient of X and Y .
r	Sample correlation coefficient, when the sample is understood.
$\varphi_X(t) = \mathbb{E}[e^{itX}]$	Characteristic function of X .
$M_X(t) = \mathbb{E}[e^{tX}]$	Moment-generating function (mgf) of X , when finite in a neighborhood of 0.
$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$	Sample mean as a statistic of a random sample $X_1, \dots, X_n$.
$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$	Observed sample mean for data $x_1, \dots, x_n$.
$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$	Sample variance as a statistic.
$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$	Observed sample variance.
$S = \sqrt{S^2}$, $s = \sqrt{s^2}$	Sample standard deviation as a statistic and its observed value.
$\hat{\theta}$	Estimator or estimate of a parameter θ , according to context.
$\hat{p}$	Estimator or estimate of a probability parameter p ; commonly the sample proportion.
$\text{SE}(\hat{\theta})$	Standard error of an estimator $\hat{\theta}$.
$L(\theta; \mathbf{x})$	Likelihood function for parameter θ given observed data $\mathbf{x}$.
$\ell(\theta; \mathbf{x}) = \ln L(\theta; \mathbf{x})$	Log-likelihood function.
H_0	Null hypothesis.
H_1 , H_a	Alternative hypothesis; H_1 is preferred.
α	Significance level; an upper bound on the Type I error probability under the specified test.
β	Type II error probability under a specified alternative.

Notation	Meaning
ν	Degrees of freedom.
$X \sim \text{Bern}(p)$	Bernoulli distribution with success probability p .
$X \sim \text{Bin}(n, p)$	Binomial distribution.
$X \sim \text{Geom}(p),$ $\mathbb{P}(X = k) = (1 - p)^{k-1}p$	Geometric distribution on $\mathbb{N}$, counting trials up to and including the first success.
$X \sim \text{Pois}(\lambda)$	Poisson distribution with parameter λ .
$X \sim \text{Unif}(a, b)$	Continuous uniform distribution on $[a, b]$.
$X \sim \text{Exp}(\lambda)$	Exponential distribution with rate λ .
$X \sim \Gamma(\alpha, \beta)$	Gamma distribution with shape $\alpha > 0$ and rate $\beta > 0$; $f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ for $x > 0$.
$X \sim \Gamma(\alpha, \theta)$	Gamma distribution with shape $\alpha > 0$ and scale $\theta > 0$; $f_X(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}$ for $x > 0$, with $\theta = 1/\beta$.
$X \sim \mathcal{N}(\mu, \sigma^2),$ $X \sim N(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2 .
$X \sim \chi_\nu^2$	Chi-squared distribution with ν degrees of freedom.
$X \sim t_\nu$	Student's t distribution with ν degrees of freedom.
$X \sim F_{\nu_1, \nu_2}$	F distribution with numerator and denominator degrees of freedom ν_1, ν_2 .
$X_n \xrightarrow{\text{a.s.}} X$	Almost-sure convergence.
$X_n \xrightarrow{\mathbb{P}} X$	Convergence in probability.
$X_n \xrightarrow{L^p} X$	Convergence in L^p .
$X_n \xrightarrow{d} X, X_n \Rightarrow X$	Convergence in distribution.

9 Geometry

In triangle notation, A, B, C denote vertices and a, b, c the opposite side lengths. Lowercase letters such as ℓ, m, n may denote lines. The notation $\overrightarrow{AB}$ may denote either the ray from A through B or the vector from A to B , according to context. In differential geometry, the Frenet frame follows the standard uppercase convention $\mathbf{T}, \mathbf{N}, \mathbf{B}$, typeset in bold to emphasize that these are vectors; uppercase italic letters such as X, Y may denote vector fields in the usual field-specific convention.

Notation	Meaning
$\triangle ABC$	Triangle with vertices A, B, C .
$\angle ABC$	Geometric angle with vertex B .
$m\angle ABC$	Measure of the angle $\angle ABC$.
$\overline{AB}$	Segment joining A and B .
AB	Length of $\overline{AB}$.
$\overrightarrow{AB}$	Ray from A through B , or vector from A to B , according to context.
$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$	Vector from A to B when $\mathbf{a}$ and $\mathbf{b}$ are the corresponding position vectors.
$\overleftrightarrow{AB}$	Line through A and B .
$\ell \parallel m$	Lines ℓ and m are parallel.
$\ell \perp m$	Lines ℓ and m are perpendicular.
$\triangle ABC \cong \triangle DEF$	Congruent triangles; corresponding vertices are listed in the same order.

Notation	Meaning
$\triangle ABC \sim \triangle DEF$	Similar triangles; corresponding vertices are listed in the same order.
$[ABC]$	Area of $\triangle ABC$.
$C(O, r)$	Circle with center O and radius r .
$\odot O$	Circle centered at O , especially in elementary geometry when the radius is understood.
$\widehat{AB}, \widehat{ACB}$	Arcs specified by their endpoints, with an additional point used when needed to distinguish the intended arc.
O, I, H, G	Circumcenter, incenter, orthocenter, and centroid of a triangle, respectively.
R, r	Circumradius and inradius of a triangle, respectively.
$s = \frac{a+b+c}{2}$	Semiperimeter of a triangle with side lengths a, b, c .
$d(P, Q)$	Distance between the points P and Q .
$\text{dist}(P, S)$	Distance from the point P to the set S .
$\text{length}(C), \ell(C)$	Length of C .
$\text{area}(S)$	Area of S .
$\text{vol}(S)$	Volume of S .
$\text{diam}(S)$	Diameter of S .
$\text{conv}(S)$	Convex hull of S .
$\text{aff}(S)$	Affine hull of S .
$\mathbf{x} = \mathbf{p} + t\mathbf{v}$	Parametric equation of a line through $\mathbf{p}$ in the direction $\mathbf{v}$.
$\mathbf{n} \cdot (\mathbf{x} - \mathbf{p}) = 0$	Hyperplane through $\mathbf{p}$ with normal vector $\mathbf{n}$.
$\mathbb{R}^n$	n -dimensional real Euclidean space.
$S^n = \{\mathbf{x} \in \mathbb{R}^{n+1} \mid \ \mathbf{x}\ = 1\}$	Unit n -sphere.
$S_r^{n-1}(\mathbf{a})$	Sphere of radius r centered at $\mathbf{a} \in \mathbb{R}^n$.
$\text{Isom}(X)$	Isometry group of the metric or geometric space X .
$\mathbb{RP}^n$	Real projective n -space.
$\mathbb{CP}^n$	Complex projective n -space.
$[x_0 : \cdots : x_n]$	Homogeneous coordinates in projective space.
$\mathbb{H}^n$	n -dimensional hyperbolic space.
$d_{\mathbb{H}}$	Hyperbolic distance, when the hyperbolic metric is understood.
$\gamma: I \rightarrow \mathbb{R}^n$	Parametrized curve, where $I \subseteq \mathbb{R}$ is an interval.
s	Arc-length parameter, when so specified.
$\mathbf{T}, \mathbf{N}, \mathbf{B}$	Unit tangent, principal normal, and binormal vectors of the Frenet frame.
κ	Curvature of a regular curve.
τ	Torsion of a space curve.
$T_p M$	Tangent space of the manifold or regular surface M at p .
$\mathbf{n}: M \rightarrow S^2$	Gauss map of an oriented surface $M \subseteq \mathbb{R}^3$, assigning its unit normal vector.
I, II	First and second fundamental forms of a regular surface.
κ_1, κ_2	Principal curvatures of a regular surface.
$K = \kappa_1 \kappa_2$	Gaussian curvature of a regular surface.
$H = \frac{\kappa_1 + \kappa_2}{2}$	Mean curvature of a regular surface under the average convention.
(M, g)	Riemannian manifold M with Riemannian metric g .
g_p	Riemannian inner product on $T_p M$.

Notation	Meaning
$\nabla_X Y$	Covariant derivative of the vector field Y in the direction of the vector field X ; in Riemannian geometry, ∇ commonly denotes the Levi–Civita connection.
Γ_{ij}^k	Christoffel symbols of a connection in local coordinates.
$\exp_p: T_p M \rightarrow M$	Riemannian exponential map at p , on its domain of definition.
$\nabla_{\dot{\gamma}} \dot{\gamma} = 0$	Geodesic equation for an affinely parametrized curve γ .

10 Number Theory

Notation	Meaning
$a \mid b$	a divides b .
$a \nmid b$	a does not divide b .
$\gcd(a, b)$	Greatest common divisor, taken nonnegative.
$\text{lcm}(a, b)$	Least common multiple.
$p^k \parallel n$	p^k exactly divides n : $p^k \mid n$ and $p^{k+1} \nmid n$; equivalently, $v_p(n) = k$.
$v_p(n), \nu_p(n)$	p -adic valuation of n ; $v_p(n)$ is preferred.
$a \equiv b \pmod{n}$	a is congruent to b modulo n .
$[a]_n$	Residue class of a modulo n .
$\mathbb{Z}/n\mathbb{Z}$	Ring of residue classes modulo n .
$(\mathbb{Z}/n\mathbb{Z})^\times$	Multiplicative group of units modulo n .
$a^{-1} \pmod{n}$	Multiplicative inverse of a modulo n .
$\text{ord}_n(a)$	Multiplicative order of a modulo n .
$\mathbb{F}_q, \text{GF}(q)$	Finite field with q elements, where q is a prime power; $\mathbb{F}_q$ is preferred.
$\left(\frac{a}{p}\right)$	Legendre symbol, for an odd prime p .
$\left(\frac{a}{n}\right)$	Jacobi symbol, for an odd positive integer n .
$\varphi(n)$	Euler totient function.
$\sum_{d \mid n}$	Sum over the positive divisors d of n .
$\prod_{p \mid n}$	Product over the prime divisors p of n .
$\sigma_k(n) = \sum_{d \mid n} d^k$	Sum of the k th powers of the positive divisors of n .
$\tau(n) = \sigma_0(n)$	Number of positive divisors of n .
$\sigma(n) = \sigma_1(n)$	Sum of positive divisors of n .
$\mu(n)$	Möbius function.
$\omega(n)$	Number of distinct prime divisors of n .
$\Omega(n)$	Number of prime factors of n , counted with multiplicity.
$\text{rad}(n)$	Product of the distinct prime divisors of n .
$(f * g)(n) = \sum_{d \mid n} f(d)g(n/d)$	Dirichlet convolution of arithmetic functions f and g .
$\pi(x)$	Number of primes not exceeding x .
$[a_0; a_1, a_2, \dots]$	Simple continued fraction.
$[a_0; a_1, \dots, a_n] = \frac{p_n}{q_n}$	n th convergent of a continued fraction.
$\Phi_n(x)$	n th cyclotomic polynomial.

Notation	Meaning
$\mathbb{Z}[i]$	Ring of Gaussian integers.
$K = \mathbb{Q}(\sqrt{d})$	Quadratic field, with squarefree $d \in \mathbb{Z}$ and $d \neq 1$.
$\mathcal{O}_K$	Ring of integers of the number field K .
$N(\mathfrak{a}) = \mathcal{O}_K/\mathfrak{a} $	Absolute norm of a nonzero integral ideal $\mathfrak{a} \subseteq \mathcal{O}_K$.
$\text{disc}(K)$	Discriminant of the number field K .
$\text{Cl}(K)$	Ideal class group of the number field K .
$h_K = \text{Cl}(K) $	Class number of the number field K .

11 Algebra

G, H usually denote groups, R, S rings, I, J ideals, M, N modules, and F, K, L fields. Modules are left modules unless otherwise stated.

Notation	Meaning
G	Group.
$e, 1_G$	Identity element of a group.
g^{-1}	Inverse of $g \in G$.
$ G $	Order (cardinality) of G .
$\text{ord}(g), g $	Order of the element g .
$\langle S \rangle$	Subgroup generated by $S \subseteq G$.
$\langle g \rangle$	Cyclic subgroup generated by g .
$H \leq G$	H is a subgroup of G .
$H \trianglelefteq G$	H is a normal subgroup of G .
gH, Hg	Left and right cosets of H in G .
$[G : H]$	Index of H in G .
G/H	Quotient group, for $H \trianglelefteq G$.
$\ker \varphi$	Kernel of the homomorphism φ .
$\text{im } \varphi$	Image of the homomorphism φ .
$\text{Aut}(G)$	Automorphism group of G .
$\text{Inn}(G)$	Inner automorphism group of G .
$Z(G)$	Center of G .
$C_G(A), C_G(x)$	Centralizer in G of $A \subseteq G$ or $x \in G$.
$N_G(A), N_G(H)$	Normalizer in G of $A \subseteq G$ or $H \leq G$.
$[x, y] := x^{-1}y^{-1}xy$	Commutator of x and y .
$[G, G], G'$	Commutator (derived) subgroup of G .
$\text{Syl}_p(G)$	Set of Sylow p -subgroups of G .
n_p	Number of Sylow p -subgroups of G .
$G \curvearrowright X$	Action of G on X .
$g \cdot x$	Action of $g \in G$ on $x \in X$.
$G \cdot x$	Orbit of x under the action of G .
$\text{Stab}_G(x), G_x$	Stabilizer of x in G .
$G \times H$	Direct product of groups.
$N \rtimes_{\varphi} H$	Semidirect product determined by $\varphi: H \rightarrow \text{Aut}(N)$, with N the normal factor.
$\langle S \mid R \rangle$	Group presentation with generators S and relations R .
C_n	Cyclic group of order n .
D_n	Dihedral group of symmetries of a regular n -gon; $ D_n = 2n$.
$S_n, \mathfrak{S}_n$	Symmetric group on n symbols.

Notation	Meaning
$A_n, \mathfrak{A}_n$	Alternating group on n symbols.
V_4	Klein four-group.
Q_8	Quaternion group of order 8.
$GL_n(F)$	General linear group over F .
$SL_n(F)$	Special linear group over F .
R	Ring.
$0_R, 1_R$	Additive and multiplicative identities of R , when disambiguation is useful.
$R^\times$	Group of units of R .
I, J	Ideals of R , when the ambient ring is clear.
$\mathfrak{p}, \mathfrak{q}$	Prime ideals.
$\mathfrak{m}, \mathfrak{n}$	Maximal ideals.
(a)	Principal ideal generated by a .
$(a_1, \dots, a_n)$	Ideal generated by $a_1, \dots, a_n$.
$I + J$	Sum of the ideals I and J .
IJ	Product of the ideals I and J .
R/I	Quotient ring.
$R[x], R[x_1, \dots, x_n]$	Polynomial rings over R in one or several variables.
$\deg f$	Degree of the polynomial f .
$M_n(R)$	Ring of $n \times n$ matrices over R .
$R[G]$	Group ring of G over R .
$\text{Frac}(R)$	Fraction field of the integral domain R .
$\text{char } R$	Characteristic of R .
$\mathbb{H}$	Quaternion division algebra over $\mathbb{R}$.
M, N	Left R -modules, unless otherwise stated.
${}_R M, M_R$	Left and right R -modules, when the side is emphasized.
M/N	Quotient module, for a submodule N of M .
$M \oplus N$	Direct sum of modules.
$\bigoplus_{i \in I} M_i$	Direct sum of an indexed family of modules.
R^n	Free R -module of rank n .
$\text{Hom}_R(M, N)$	Set of R -module homomorphisms from M to N .
$\text{End}_R(M)$	Endomorphism ring of the R -module M .
$\text{Aut}_R(M)$	Automorphism group of the R -module M .
$\text{Ann}_R(M)$	Annihilator of the R -module M .
$M_R \otimes_R {}_R N, M \otimes_R N$	Tensor product over R ; the first notation emphasizes that M is a right R -module and N a left R -module. The side distinction is normally suppressed when R is commutative or the required module structures are otherwise clear.
K/F	Field extension of F by K .
$[K : F]$	Degree of the field extension K/F .
$F(\alpha), F(\alpha_1, \dots, \alpha_n)$	Fields generated over F by the indicated elements.
$F[\alpha]$	Subring (equivalently, F -subalgebra) generated by F and α .
$m_{\alpha, F}(x)$	Minimal polynomial of α over F .
$\overline{F}$	Algebraic closure of F .
$\text{Gal}(L/K)$	Galois group of L/K .
L^H	Fixed field of $H \leq \text{Gal}(L/K)$.
$\text{Tr}_{L/K}(\alpha)$	Field trace of $\alpha \in L$ from L to K .

Notation	Meaning
$N_{L/K}(\alpha)$	Field norm of $\alpha \in L$ from L to K .
$\text{trdeg}_K L$	Transcendence degree of L over K .

12 Topology

Notation	Meaning
$(X, \mathcal{T})$	Topological space X with topology $\mathcal{T}$.
$\mathcal{T}$	Topology on the underlying set X , when X is understood.
$\mathcal{B}$	Basis for a topology.
$\mathcal{S}$	Subbasis for a topology.
A° , $\text{int } A$	Interior of A .
$\overline{A}$, $\text{cl } A$	Closure of A .
∂A	Boundary of A .
(X, d)	Metric space X with metric d .
$\mathcal{T}_A = \{A \cap U \mid U \in \mathcal{T}\}$	Subspace topology on $A \subseteq X$.
$\pi_i: \prod_{j \in J} X_j \rightarrow X_i$	Coordinate projection of a product space, for $i \in J$.
$X/\sim$	Quotient space when the quotient set is equipped with the quotient topology.
$q: X \twoheadrightarrow X/\sim$	Canonical quotient map.
X/A	Quotient space obtained by collapsing the subspace $A \subseteq X$ to a point.
$C(X, Y)$	Set of continuous maps from X to Y .
$X \cong Y$	Homeomorphic spaces.
$\text{Homeo}(X)$	Group of self-homeomorphisms of X .
$\iota: A \hookrightarrow X$	Inclusion map of a subspace $A \subseteq X$.
$f: X \hookrightarrow Y$	Topological embedding when explicitly stated; the hook arrow by itself retains its general meaning of injectivity.
(X, x_0)	Pointed space with basepoint x_0 .
$I = [0, 1]$	Unit interval, especially for paths and homotopies.
$\gamma: I \rightarrow X$	Path in X .
$\gamma(0) = \gamma(1) = x_0$	Loop based at x_0 .
$H: X \times I \rightarrow Y$	Homotopy between maps $X \rightarrow Y$.
$f \simeq g$	Homotopic maps $f, g: X \rightarrow Y$.
$X \simeq Y$	Homotopy equivalent spaces.
$[\gamma]$	Homotopy class of a path or loop γ , according to context.
$\pi_1(X, x_0)$	Fundamental group of X based at x_0 .
$\pi_n(X, x_0)$	n th homotopy group of the pointed space (X, x_0) , for $n \geq 1$.
f_*	Homomorphism induced by a map f on a covariant invariant such as homotopy or homology, with the invariant clear from context.
f^*	Homomorphism induced by a map f on a contravariant invariant such as cohomology, with the invariant clear from context.
$p: \tilde{X} \rightarrow X$	Covering map, with $\tilde{X}$ a covering space of X .
$\tilde{X}$	Covering space; it denotes a universal cover only when explicitly specified.
$\text{Deck}(p)$	Group of deck transformations of the covering p .
$D^n = \{\mathbf{x} \in \mathbb{R}^n \mid \ \mathbf{x}\ \leq 1\}$	Closed n -disk.

Notation	Meaning
$X \vee Y$	Wedge sum of pointed spaces X and Y .
T_0, T_1, T_2, T_3, T_4	Separation axioms; here T_2 means Hausdorff, T_3 means T_1 and regular, and T_4 means T_1 and normal.
M^n	n -dimensional topological manifold, when the dimension is displayed.
∂M	Boundary of a manifold M .
Σ_g	Closed connected orientable surface of genus g .
$M \# N$	Connected sum of manifolds M and N .
$\chi(X)$	Euler characteristic of X .
$H_n(X; G)$	n th homology group of X with coefficients in G .
$H_n(X)$	Integral homology group $H_n(X; \mathbb{Z})$ when coefficients are omitted.
$\tilde{H}_n(X; G)$	n th reduced homology group of X with coefficients in G .
$H^n(X; G)$	n th cohomology group of X with coefficients in G .
$b_n(X)$	n th Betti number; for example, $b_n(X) = \dim_{\mathbb{Q}} H_n(X; \mathbb{Q})$ when this dimension is finite.

13 Miscellaneous Notation

Notation	Meaning
$\lfloor x \rfloor$	Floor of x .
$\lceil x \rceil$	Ceiling of x .
$\{x\}$	Fractional part $x - \lfloor x \rfloor$, when unambiguous.
$\operatorname{sgn}(x)$	Sign of a real number or permutation, according to context.
$a \propto b$	a is proportional to b ; in a functional context, the proportionality factor is independent of the variable under discussion.
Δq	Finite change in a quantity q , especially in a discrete or applied context.
δ_{ij}	Kronecker delta: 1 if $i = j$ and 0 otherwise.
ϵ_{ijk}	Three-dimensional Levi-Civita symbol: the sign of the permutation (i, j, k) when (i, j, k) is a permutation of $(1, 2, 3)$, and 0 otherwise.